Assignment 2

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1 Introduction

This assignment investigates the robustness of different networks under progressive node removal attacks. Robustness is evaluated by monitoring the size of the giant component (GC) during node removal. A network is considered robust if the GC remains large even after many nodes are removed.
Three networks are analyzed:

- an Erdős–Rényi random graph (ER);
- a Barabási–Albert scale-free graph (BA);
- a network built from the MyAnimeList dataset.

Four attack strategies are applied:

- random node removal;
- highest degree node removal;
- highest PageRank node removal;
- highest betweenness node removal (approximate).

Finally, different edge-addition strategies are applied to the anime network in order to evaluate possible improvements in robustness.

2 Attack Methodology

The attack procedure is iterative:

1. compute the largest connected component;
2. select nodes according to the attack strategy;
3. remove the selected nodes;
4. recompute the largest component.

The robustness measure is:

$$\frac{G}{N}$$

where G is the size of the giant component after removal and N is the initial number of nodes.
The collapse point is defined as the moment when the giant component falls below 50% of its original size.

Technical Decisions

Several technical decisions were required to ensure correctness and computational feasibility:

- **Static attacks**: degree, PageRank and betweenness rankings are computed once on the initial graph and reused during the attack.
- **Approximate betweenness**: betweenness centrality is computed using sampling to reduce computational cost.
- **Batch removal**: for all networks, nodes are removed in fixed-size batches at each iteration.
- **Giant component tracking**: after each removal, only the largest connected component is considered.

These decisions reduce computation time while preserving correctness and ensuring that attack curves are not distorted.

3 Experiments on Synthetic Networks

3.1 Erdős–Rényi Network

The ER graph ($N = 1000$, $p = 0.01$) shows homogeneous robustness. Random failures and targeted attacks produce identical behavior because the degree distribution is relatively uniform.

Collapse points:

- Random: 500 nodes removed (50.0%),
- Degree: 500 nodes removed (50.0%),
- PageRank: 500 nodes removed (50.0%),
- Betweenness: 500 nodes removed (50.0%).

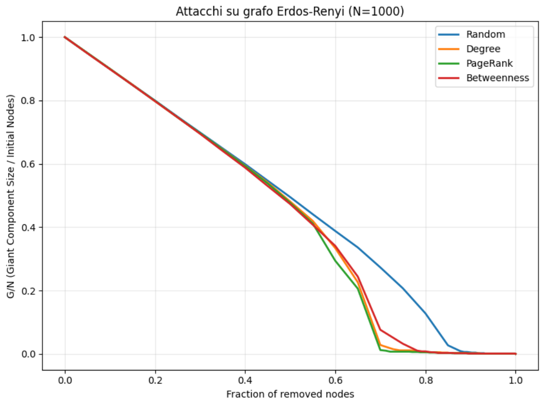

Figure 1: Attack results on the Erdős–Rényi graph. All curves collapse at 50%.

The identical collapse point for all strategies indicates that no attack strategy has a significant advantage. Since the ER graph has a homogeneous degree distribution, there are no hubs whose removal disproportionately affects connectivity.

3.2 Barabási–Albert Network

The BA graph ($N = 1000$, $m = 5$) shows the typical behavior of scale-free networks. Random failures have limited effect, while targeted attacks are significantly more damaging because they remove highly connected hubs.

Collapse points:

- Random: 500 nodes removed (50.0%),
- Degree: 375 nodes removed (37.5%),
- PageRank: 350 nodes removed (35.0%),
- Betweenness: 375 nodes removed (37.5%).

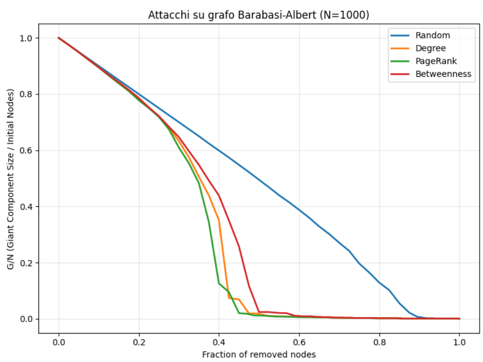

Figure 2: Attack results on the Barabási–Albert graph. Targeted attacks collapse the network earlier.

Targeted attacks cause earlier collapse because the BA network contains hubs that are essential for global connectivity. PageRank is the most effective strategy, as high-PageRank nodes tend to be both highly connected and structurally central.

3.3 Watts–Strogatz Network

The WS graph is generated with 1000 nodes, where each node is initially connected to its 10 nearest neighbors ($k = 10$), and each edge is rewired with probability $p = 0.1$. This produces a small-world network with high local clustering and relatively short path lengths.
Collapse points:

- Random: 500 nodes removed (50.0%),
- Degree: 450 nodes removed (45.0%),
- PageRank: 500 nodes removed (50.0%),
- Betweenness: 500 nodes removed (50.0%).

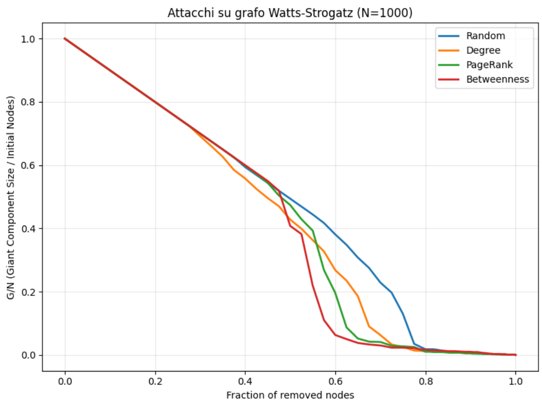

Figure 3: Attack results on the Watts-Strogatz graph.

Random, PageRank and Betweenness attacks behave similarly, indicating that these strategies do not isolate a small set of structurally critical nodes. Degree attack is slightly more effective because removing high-degree nodes disrupts local clusters more quickly.

4 Anime Similarity Network

The anime similarity network contains:

- 3579 nodes,
- 780393 edges,
- density = 0.1219,
- average degree $\approx$ 436.10.

The network is dense and highly clustered, but not fully connected: there are 728 connected components. The giant component contains 2838 nodes (79.3% of the network).
Collapse points:

- Random: 1046 nodes removed (29.2%),
- Degree: 938 nodes removed (26.2%),
- PageRank: 1055 nodes removed (29.5%),
- Betweenness: 966 nodes removed (27.0%).

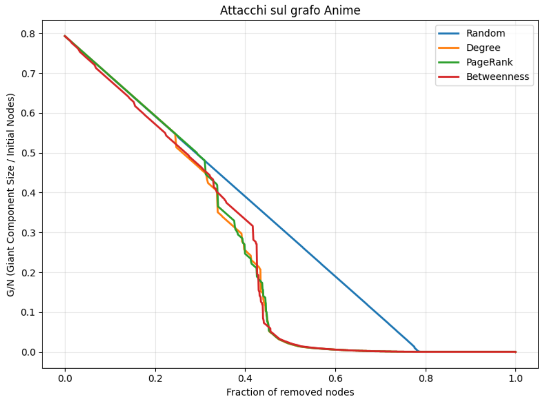

Figure 4: Attack results on the anime similarity network. Degree and betweenness attacks are the most damaging.

The collapse threshold represents the fraction of nodes that must be removed before the network reaches its defined collapse condition.
A lower collapse threshold indicates greater vulnerability, as fewer node removals are required to significantly disrupt the network.
In this case:

$$Random\ :0.2923 \qquad Degree\ :0.2621 \qquad PageRank\ :0.2948 \qquad Betweenness\ :0.2699$$

Therefore, the network is most vulnerable to degree-based attacks and most resilient to PageRank-based attacks. Although PageRank-based attacks may affect the network structure differently during the removal process, they require the largest fraction of removed nodes before reaching the defined collapse threshold.

PageRank-based attacks are less effective than Degree or Betweenness attacks because PageRank tends to select nodes that are central within dense genre-based clusters but are not necessarily critical for maintaining global connectivity. In contrast, Degree and Betweenness attacks target highly connected hubs and bridging nodes, whose removal fragments the network more effectively.

5 Building Robustness

To evaluate how structural modifications affect robustness, three edge-addition strategies were applied to the anime similarity network. Each strategy adds new edges according to a different structural criterion:

1. **Ring Degree**: connects neighbors of the top-3 highest-degree nodes, adding 600 new edges.
2. **Ring Betweenness**: connects neighbors of the top-3 highest-betweenness nodes, adding 1,487,668 new edges.
3. **High Clustering**: connects neighbors of nodes with the highest clustering coefficient, adding 200 new edges.

These strategies differ significantly in cost (number of added edges) and in the type of structural reinforcement they introduce.

5.1 Critical Threshold

The theoretical critical threshold for random failures is given by the Molloy–Reed criterion:

$$f_c = 1 - \frac{1}{\frac{k_2}{k_1} - 1},$$

where k_1 and k_2 are respectively the first and second moments of the degree distribution. Higher values of f_c indicate greater robustness under random node removal.

The results obtained after applying each robustness strategy are reported in Table 1.

Graph	Number of nodes	Number of edges	k_1	k_2	f_c
Original	3579	780393	436.10	351698.2	0.998758
Ring Degree	3579	780993	436.43	352254.1	0.998760
Ring Betweenness	3579	2268061	1267.43	2577432.0	0.999508
High Clustering	3579	780593	436.21	351857.1	0.998759

Table 1: Critical threshold comparison after robustness strategies.

The anime network is already extremely robust ($f_c \approx 0.9988$), due to its high density and large average degree. As a consequence, strategies that add a small number of edges (Ring Degree and High Clustering) produce only marginal improvements in f_c.

The Ring Betweenness strategy yields the largest increase in robustness, substantially raising both k_1 and k_2. However, this improvement comes at a very high cost: more than 1.4 million new edges are added. This highlights a clear trade-off between structural reinforcement and modification cost.

6 Conclusion

The analysis shows that network robustness depends strongly on both structural properties and the attack strategy applied.
Random failures are generally less harmful because they do not specifically target structurally important nodes.

The anime similarity network (Figure 4) demonstrates high robustness thanks to its dense structure, large giant component, and strong redundancy. Targeted attacks are more effective than random removal, but the network maintains substantial resilience even when important nodes are removed.

The robustness-enhancing strategies modify the network in different ways, the results highlight how network topology shapes robustness and how targeted interventions can influence resilience in complex systems. In dense networks such as the anime similarity graph, meaningful improvements in robustness require substantial structural modifications, illustrating the trade-off between cost and effectiveness in network design.